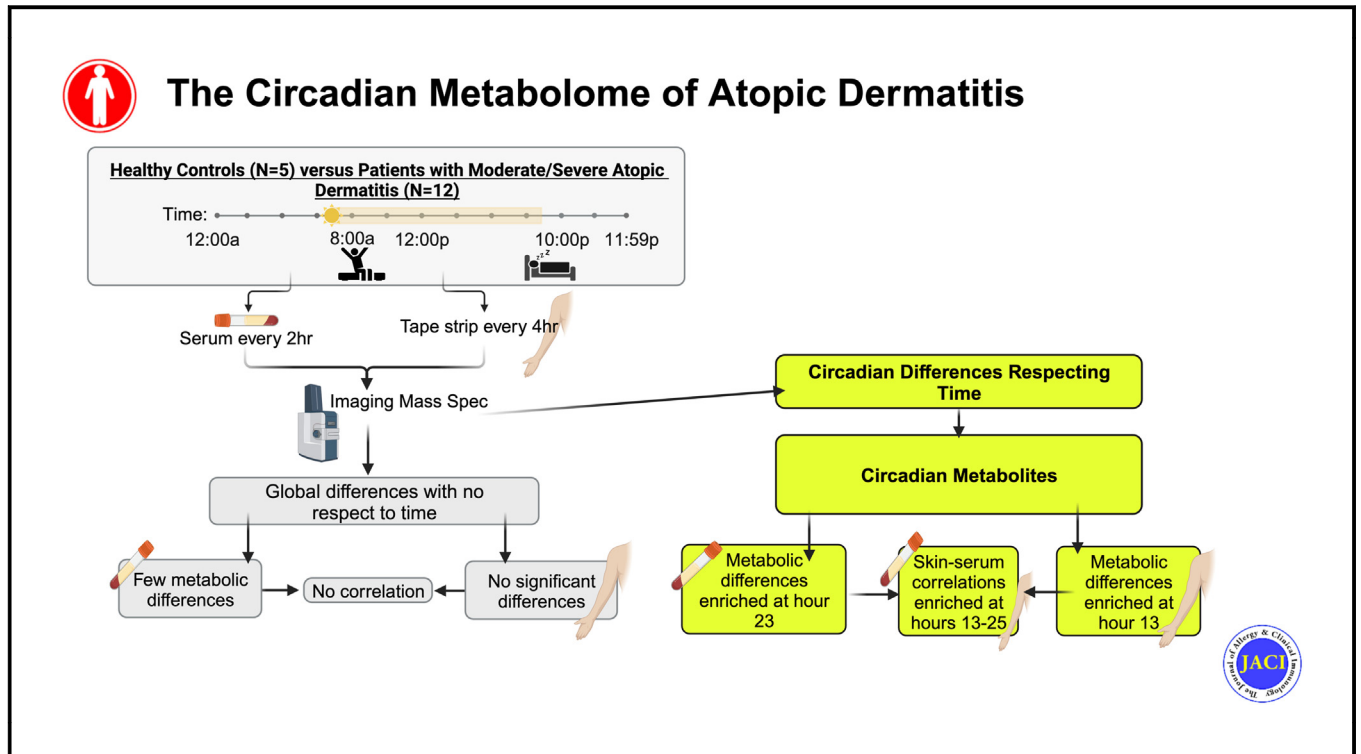


The circadian metabolome of atopic dermatitis



Grace Ratley, BSPH, Jordan Zeldin, BS, Prem Prashant Chaudhary, PhD, Manoj Yadav, PhD, Amy S. Paller, MD, MS, Phyllis Zee, MD, PhD, et al

GRAPHICAL ABSTRACT



Capsule summary: An untargeted metabolomics study in atopic dermatitis identified distinct time-based metabolic changes in skin and serum compared to healthy controls, suggesting that attention to sample collection time could improve biomarker discovery.

The circadian metabolome of atopic dermatitis



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Background: Atopic dermatitis (AD) is a chronic inflammatory skin disease characterized by dry, pruritic skin. Several studies have described nocturnal increases in itching behavior, suggesting a role for the circadian rhythm in modulating symptom severity. However, the circadian rhythm of metabolites in the skin and serum of patients with AD is yet to be described.

Objective: We sought to assess circadian patterns of skin and serum metabolism in patients with AD.

Methods: Twelve patients with moderate to severe AD and 5 healthy volunteers were monitored for 28 hours in a controlled environment. Serum was collected every 2 hours and tape strips every 4 hours from both lesional and nonlesional skin in participants with AD and location-, sex-, and age-matched healthy skin of controls. We then performed an untargeted metabolomics analysis, examining the circadian peaks of metabolism in patients with AD.

Results: Distinct metabolic profiles were observed in AD versus control samples. When accounting for time of collection, the greatest differences in serum metabolic pathways were observed in arachidonic acid, steroid biosynthesis, and terpenoid backbone biosynthesis. We identified 42 circadian peaks in AD or control serum and 17 in the skin. Pathway enrichment and serum-skin metabolite correlation varied throughout the day. Differences were most evident in the late morning and immediately after sleep onset.

Conclusions: Although limited by a small sample size and observational design, our findings suggest that accounting for sample collection time could improve biomarker detection studies in AD and highlight that metabolic changes may be associated with nocturnal differences in symptom severity. (*J Allergy Clin Immunol* 2024;153:1148-54.)

Key words: Atopic dermatitis, circadian rhythm, metabolomics, bioinformatics

Abbreviation used

AD: Atopic dermatitis

INTRODUCTION

Atopic dermatitis (AD) is a chronic inflammatory skin disease characterized by dry, pruritic skin.¹ Several studies have described nocturnal increases in scratching behavior, suggesting a role for circadian rhythms in modulating symptom severity.^{2,3} Circadian rhythms underlie skin physiology, such as DNA repair, transepidermal water loss, and keratinocyte proliferation.^{4,5} Understanding the mechanisms underpinning nocturnal increases in itch could improve therapeutic precision. Furthermore, identification of rhythmic biomarkers could guide new methods for predicting and preventing flares.

Metabolomics is a powerful tool for investigating the biochemical activity of cells and tissues. A recent study of the human metabolome found that 15% of identifiable metabolites in saliva and serum fall under circadian control.⁶ Further evidence suggests that these changes may be organ-specific.⁷ Although several studies have characterized the metabolic profile of AD, especially disrupted lipid synthesis,^{8,9} the role of the circadian rhythm in regulating these biochemical pathways has not been elucidated. In addition, published studies have reported data from either blood or skin, but not both. We therefore conducted an untargeted metabolomics study of the blood serum and skin of patients with AD in a highly controlled circadian experiment over 28 hours.

RESULTS AND DISCUSSION

We performed an untargeted metabolomics analysis of serum and stratum corneum samples from skin tape strips collected from 5 healthy controls and 12 patients with AD (Table I). Time point 0 was set to 9 hours after habitual bedtime during the previous week. Serum was collected every 2 hours and tape strip samples every 4 hours from healthy antecubital fossa skin of controls and both lesional (antecubital fossae) and nonlesional (forearm) skin of patients with AD. Although any collection procedure during a sleep study may disturb participants' sleep and could confound results, we found that biomarkers of central circadian rhythm (melatonin and cortisol collected over 28 hours) were not disturbed (Fig 1, A). Samples were subject to matrix-assisted laser desorption ionization-trapped ion mobility spectrometry time-of-flight mass spectrometry.

Pooling all samples from patients with AD versus controls, we identified significant differences in the metabolic profiles of serum in patients with AD versus controls (Fig 1, B and C). Overall, 3323 peaks with unique mass-to-charge (*m/z*) ratios were detected in 228 serum samples. Collisional cross-section values,

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TABLE I. Demographic and clinical characteristics of the study population (n = 17)

Variable	Control	AD	Overall	P value
Sex				>.999*
Female	2 (28.57)	5 (71.43)	7 (100.00)	
Male	3 (30.00)	7 (70.00)	10 (100.00)	
Age (y), mean $\pm$ SD	19.89 $\pm$ 3.55	19.45 $\pm$ 2.77	19.58 $\pm$ 2.91	>.999†
EASI, mean $\pm$ SD	—	16.53 $\pm$ 13.69	16.53 $\pm$ 13.69	>.999†
Asthma				>.999*
No	4 (33.33)	8 (66.67)	12 (100.00)	
Yes	1 (20.00)	4 (80.00)	5 (100.00)	
Rhinitis				.685*
No	2 (50.00)	2 (50.00)	4 (100.00)	
Yes	3 (23.08)	10 (76.92)	13 (100.00)	

Values are n (%) unless otherwise indicated.

EASI, Eczema Area and Severity Index.

* χ^2 test.

†Fisher exact test.

which can distinguish isomers with the same *m/z* value, were used to annotate peaks with metabolite names using the collisional cross-section consortium analyte list in Metaboscape 2021b (47 metabolites identified). Furthermore, the LIPID MAPS spectral library was used to putatively annotate peaks on the basis of *m/z* values alone (170 peaks with putative annotations). Unpaired *t* tests revealed 105 peaks with significant differences in abundance in the serum of controls and patients with AD (adjusted *P* < .05; Fig 1, B); of these, 11 were annotated (Fig 1, C).

Pathway analysis conducted in MetaboAnalyst¹⁰ (freely available at metaboanalyst.co) identified significant differences in pyruvate metabolism (*P* < .05) and trends toward differences in the citric acid cycle, glycolysis, and thiamine metabolism (*P* = .09) when not accounting for collection time (Fig 1, D; see also Fig E1 and Table E1 in this article's Online Repository at www.jacionline.org).¹¹ Alterations in pyruvate metabolism have been reported in both lesional and nonlesional keratinocytes in AD and hypothesized to be related to increased oxidative stress.^{12,13} In addition, patients with AD with high IgE levels have increased serum citric and lactic acid.¹⁴ Although we did not evaluate IgE levels in participants, our data corroborate these studies, showing that differences in energy metabolism and mitochondria function may be present in AD.

When assessed by collection hour, the variability of serum metabolites was more obvious (Fig 1, E). Differences measured by analysis of similarities (ANOSIM) were more pronounced in the serum immediately before waking (hour 23) (Fig 1, F; see also Fig E2 in this article's Online Repository at www.jacionline.org). Using the MetaCycle R package (freely available at github.com/gangwug/MetaCycle), a composite of 3 algorithms (ie, ARSER, JTK_Cycle, and Lomb-Scargle), we identified 28 peaks in AD serum and 14 in control serum with a significant (*P* < .05) periodicity between 20 and 28 hours (see Fig E3, A, in this article's Online Repository at www.jacionline.org).¹⁵ These results suggest that existing studies with nonstandard collection times may be limited in their assessment of metabolic differences between patients with AD and controls.

Although the rhythmicity and differences in serum pathway enrichment did not reach statistical significance, arachidonic acid

metabolism, steroid hormone biosynthesis, and terpenoid backbone biosynthesis showed time-shifted patterns in patients with AD and controls (Fig 1, C; Fig E3, B). Terpenoids are precursor molecules to endogenous steroids. Arachidonic acid is a proinflammatory lipid released in response to injury that promotes the formation of prostaglandins and leukotrienes. Previous studies have shown that metabolites of arachidonic acid such as leukotriene B₄ and prostaglandin E₂ are elevated in AD.¹⁶ Our results show that therapies targeting these proinflammatory lipids should consider timing of dosing.

For tape strip samples, 1-way ANOVA was used to identify 302 peaks with group differences in abundance (adjusted ANOVA *P* < .05). To account for the larger file size of tape strip data across 197 samples, 104 of the 410 peaks detected were annotated using *m/z* values with xMSannotator (freely available at github.com/yufree/xMSannotator) referencing the Human Metabolome Database (see Table E2 in this article's Online Repository at www.jacionline.org).¹⁷ Unpaired *t* tests showed 295 differentially abundant peaks between lesional and nonlesional skin in patients with AD, 302 between lesional skin of patients with AD and nonlesional in controls, and 260 between nonlesional skin of patients with AD and controls. Of these, 85 were annotated (Fig 2, A). Overall similarity by ANOSIM was identified (Fig 2, B). Because of the small number of metabolites annotated by MetaboAnalyst in the tape strip samples, pathway analysis was not possible.

Using the MetaCycle R package, we identified 17 metabolites in lesional skin, but none in nonlesional or control skin (*P* < .05) (see Fig E4 in this article's Online Repository at www.jacionline.org). These metabolites appeared to be under circadian control, with a periodicity between 20 and 28 hours. As with serum (Fig 1, D), sorting skin metabolites by collection hour demonstrated more distinct clustering between controls versus patients' lesional and nonlesional skin (Fig 2, C).

Several research groups have searched for AD biomarkers using minimally invasive alternatives to skin biopsy, including blood tests, to identify inflammatory cytokines to predict AD severity.⁸ The lack of consistent metabolic markers for AD across studies has been attributed to small sample size and differences in

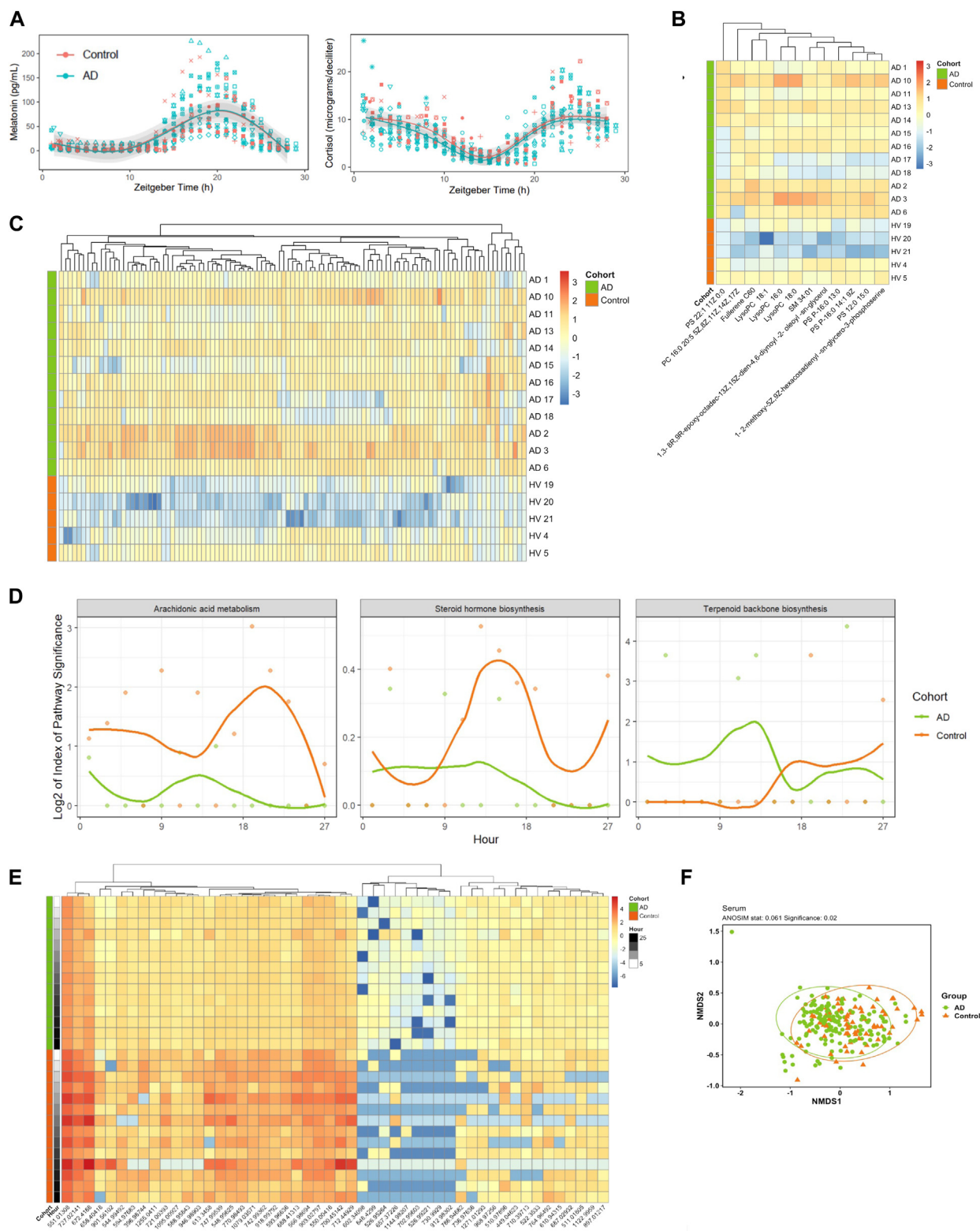


FIG 1. A, Hourly cortisol and melatonin levels of all patients collected during the 28-hour collection period. Plot of individual values by patient, color-coded by AD ($n = 12$) vs control ($n = 5$), with model-predicted values (smoothed lines) with 95% CIs as the shaded region. **B**, A total of 105 significantly different pooled serum peaks (adjusted unpaired t test $P < .05$). **C**, Eleven significantly different annotated serum metabolites (adjusted unpaired t test $P < .05$). **D**, Pathway enrichment over a day as measured by the Index of Pathway Significance value. **E**, Top 50 differentially abundant serum peaks stratified by hour (white, ZT-0; black, ZT-27). **F**, NMDS plots of pooled serum data across all time points. ANOSIM, Analysis of similarity; NMDS, non-metric multidimensional scaling.

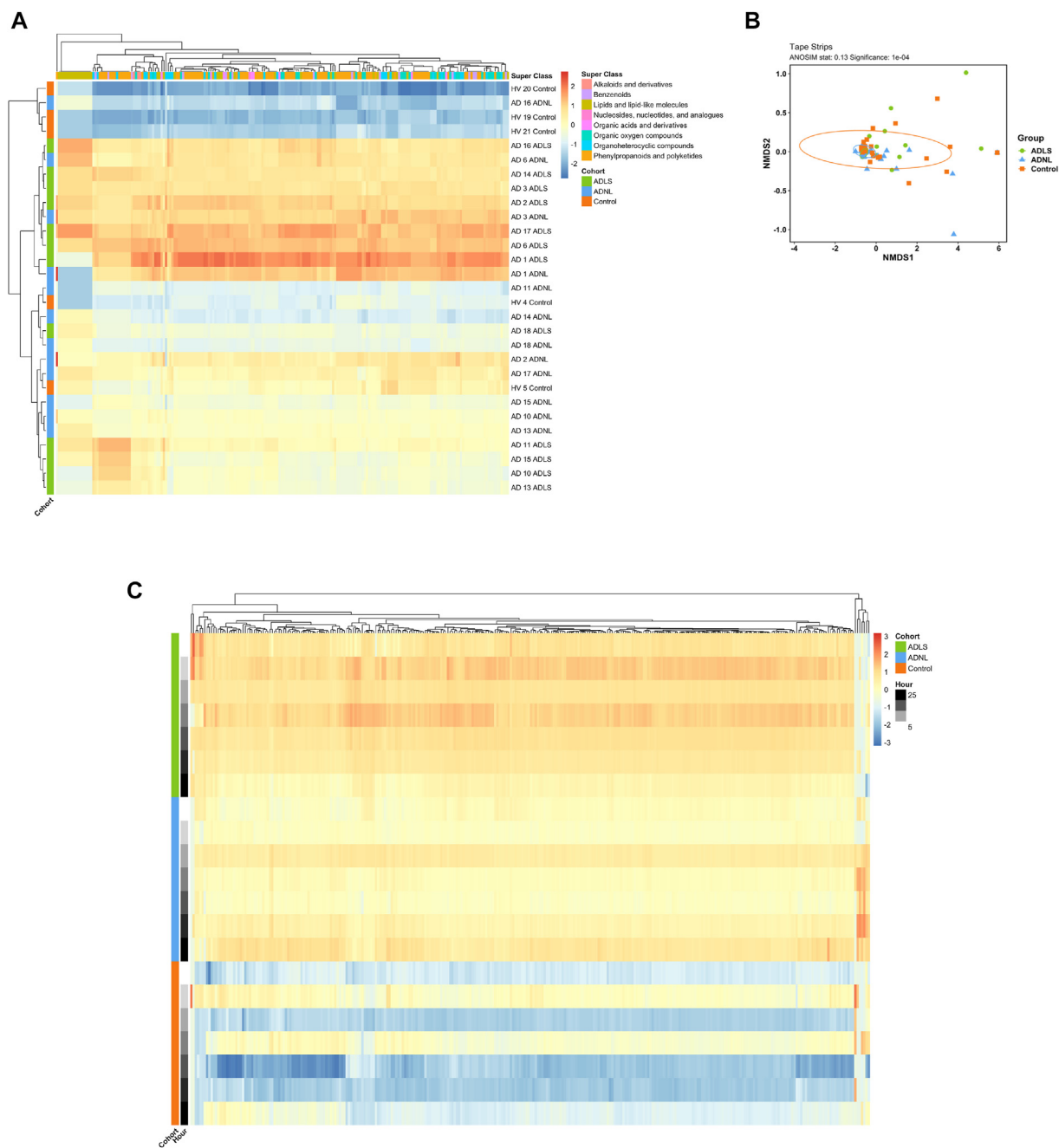


FIG 2. **A**, Eighty-five annotated metabolites significantly different between control, nonlesional (ADNL), and lesional (ADLS) skin (adjusted ANOVA $P < .05$). All possible annotations for each feature are shown. **B**, NMDS plot of pooled skin data across all time points. **C**, Top 50 differentially abundant skin metabolites stratified by hour. ADLS, Lesional skin in atopic dermatitis; ADNL, nonlesional skin in atopic dermatitis; ANOSIM, analysis of similarity; NMDS, non-metric multidimensional scaling.

patient environment and genetics. Our data suggest that sample collection time may be another factor contributing to inconsistent results.

We generated correlational matrices of the top 50 differentially abundant metabolites in the serum and the top 25 in skin (Fig 3,

A-D). As expected, metabolites of the same sample type were highly correlated at all time points. However, the predictive serum biomarkers did not correlate with the skin biomarkers when viewed in aggregate (without considering sample collection time) (Fig 3, A). In contrast, separating each sample by time of

Serum Versus Skin Metabolite Correlation

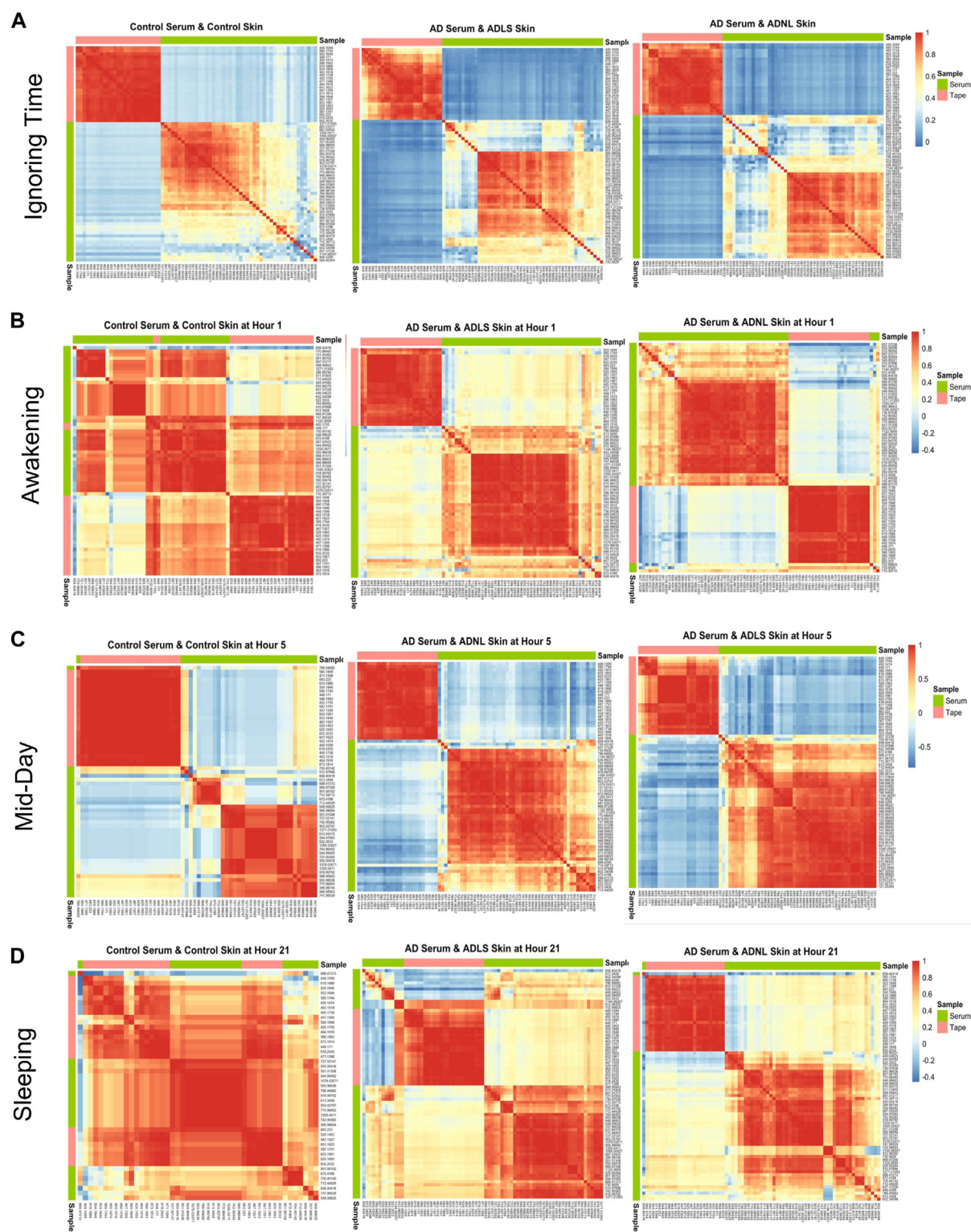


FIG 3. Correlation plots for top 50 differentially abundant peaks in serum and top 25 in skin. **A**, Pooled across all time points. **B**, At ZT-1 representing awakening metabolism. **C**, At hour 5 representing midday metabolism. **D**, At hour 21 representing sleeping metabolism. *ADLS*, Lesional skin in atopic dermatitis; *ADNL*, nonlesional skin in atopic dermatitis.

collection before evaluation revealed that the correlation between serum and skin metabolites changes over the course of a day. Serum markers of interest would be more predictive of the skin biomarker candidates in the early morning and late evening time points (Fig 3, B-D; see also Fig E5 in this article's Online Repository at www.jacionline.org).

Lasso regression identified metabolites characteristic of AD at different time points (see Figs E6 and E7 in this article's Online Repository at www.jacionline.org). In addition to skin and serum metabolites, demographic characteristics including sex, age, total sleep time, and wake-up time were added to the model in comparisons of control and AD samples. We did not identify any metabolite as a consistent marker across time points of AD in either the skin or the serum (Fig E7, D). One metabolite with an *m/z* ratio of 449.1099, putatively annotated as a sterol lipid by Metabolomics Workbench, was commonly selected by the lasso model as a discriminating feature between lesional and nonlesional skin. In a similar model, we examined the correlation of metabolites with Eczema Area and Severity Index scores and found no consistent metabolites associated with AD severity across the 27-hour sampling period. These results suggest that the metabolites and metabolic pathways that may best differentiate AD from healthy skin and serum may shift during a day.

This work, although limited by the small sample size and observational nature, offers the first circadian evaluation of the metabolome associated with AD. Further work is needed to prospectively correlate the metabolites indicated in our analysis with predictive models of AD onset and severity. This work may be further enhanced by assessment of the circadian patterns of established markers of allergic inflammation (such as CCL17 or IL-4) and correlation with the metabolomics presented in this work and accessible under MetaboLights accession number MTBLS9508.¹⁸ In addition, techniques such as tandem mass spectrometry could be used to further identify circadian metabolites in samples. Overall, our findings suggest that consideration of sample collection time could improve the biomarker detection efforts for clinical studies in AD and highlight that nocturnal differences in symptom severity might be associated with targetable changes in endogenous steroids related to terpenoid metabolism and proinflammatory lipid mediators such as arachidonic acid.

DISCLOSURE STATEMENT

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Disclosure of potential conflict of interest: The authors declare that they have no relevant conflicts of interest.

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Key messages

- Accounting for sample collection time could improve precision of clinical study findings.
- Nocturnal differences in AD symptom severity might be associated with metabolic disconnects between skin and serum.

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